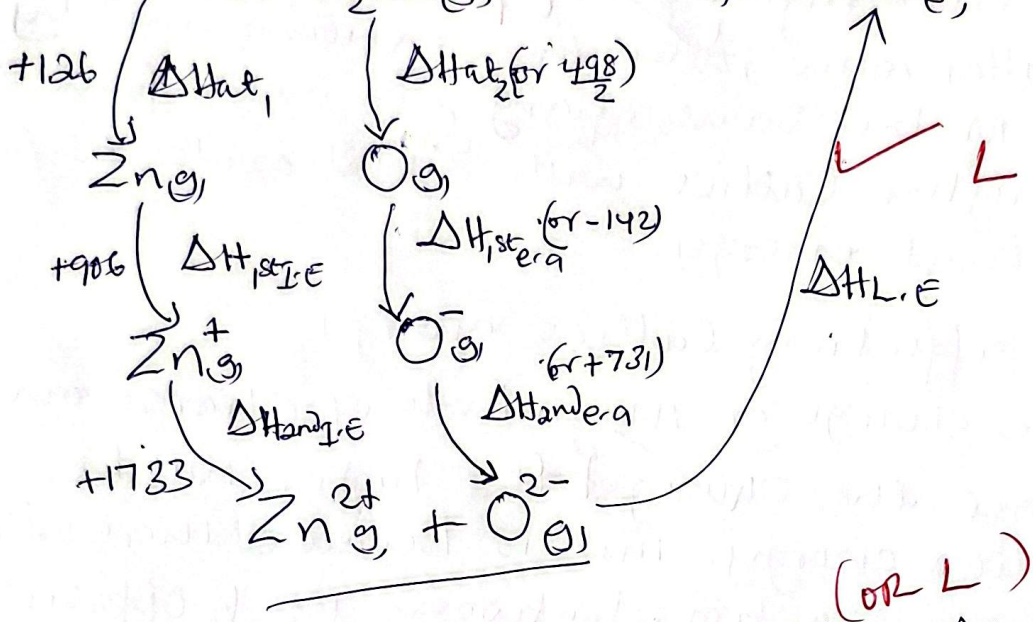
ITEM 1

Equation for formation of zinc oxide

$$Zn_{(s)} + \tfrac{1}{2}O_{2(g)} \xrightarrow{\Delta H_f} ZnO_{(s)}$$



$$\Delta H_f = \Delta H_{at_1} + \Delta H_{1st\ I.E} + \Delta H_{2nd\ I.E} + \Delta H_{at_2} + \Delta H_{1st\ e.a} + \Delta H_{2nd\ e.a} + \Delta H_{L.E} \quad (\text{or } L)$$

$$= +126 + 906 + 1733 + \left(\frac{498}{2}\right) + (-142) + 731 + (-4010)$$

$$= -407\ KJ\ mol^{-1}$$

(Sign must be there)

$$ZnO_{(s)} + M_{(s)} \longrightarrow MO_{(s)} + Zn_{(s)}$$

$$\Delta H_r = \Sigma \Delta H_f(\text{products}) - \Sigma \Delta H_f(\text{reactants}) \quad (\text{or } L)$$

$$= (-602) - (-407) = -195\ KJ\ mol^{-1}$$

$$= -195000\ J\ mol^{-1}$$

$$\Delta G = -nEF \quad (\text{or } L)$$

$$Zn^{2+}_{(s)} + M_{(s)} \longrightarrow M^{2+}_{(s)} + Zn_{(s)}$$

$$-195000 = 2 \times E \times 96500$$

$$E = \frac{-195000}{2 \times 96500} = +1.01\ V$$

E.m.f for the reaction $= +1.01\ V$ and is

less than what is required for the learners electrochemical cell.

Therefore they cannot use metal M but instead should use another metal which can form cation with higher charge and smaller ionic radii or smaller ionic radii than cation formed by metal M but same charge so as to form its form higher lattice with metal oxides to produce more heat energy.

Factors affecting lattice energy.

(i) ionic charge (or magnitude of charge on ions)

The bigger the charge the higher will be the lattice energy. This is because there will be strong attraction between the oppositely charged ions hence producing more energy

Ionic radii

The smaller the ionic radii the higher will be lattice energy. This is because the ions will approach each other closely and attract each other firmly resulting into release of greater amount of heat energy.

Criteria	ITEM 1	Scores
2L + 2Fe		= 2
2La + 1Fe		= 1
3Da + Di + Dc		= 3
3Da + Di		= 2
(2/3) 2 Da		= 1
Ca + Cr		= 2
Ca / Cr		= 1
TOTAL		7

ITEM 2

(a) The aromatic organic compound is immiscible with water. When the steam is passed through a liquid mixture containing the aromatic organic compound both the water and organic compound vaporise independtly such that at any given temperature the total vapour pressure above the liquid mixture is equal to the sum of the two vapour pressures. Less amount of heat energy therefore will be required to make the vapour pressure of the liquid mixture equal to the outside pressure in order for the liquid mixture to boil. This makes the liquid mixture to boil at a temperature below the boiling point of the aromatic organic compound. Consequently this enables the aromatic organic compound to be distilled off at a temperature below its boiling point at which can decompose.

Temp /°C	92	94	96	98	100
V.P of organic substance/KPa	6	8	12	15	17
V.P of water/KPa	74	80	88	94	101
Total V.P /KPa	80	88	100	109	118

Steam is passed through the liquid mixture which has slightly been warmed. The vapour of the organic compound and water is given off. With increasing temperature the vapour pressure builds up and at the temperature when the total vapour pressure of the liquid mixture equals to that of external pressure or atmospheric pressure the liquid mixture boils. The vapour is condensed and a distillate is obtained

which contains both water and the organic compound. Da The water is separated from the organic compound using a separating funnel. The small traces of water Da are removed using drying agent such as anhydrous calcium chloride.

(ii) $\frac{V.P_w \times R.f.M_w}{V.P_o \times R.f.M_o} = \frac{\text{Amount of water in the distillate}}{\text{Amount of organic in the distillate}}$ L

$V.P_w = 89\ kPa$. $V.P_o = 12\ kPa$

$$\frac{89 \times 18}{12 \times R.f.M_o} = \frac{1.6}{1.1}$$ Da

$$\Rightarrow R.f.M_o = \frac{(89 \times 18 \times 1.1)}{(12 \times 1.6)} = 91.78$$

% of hydrogen: $100 - 77.42 - 15.05 = 7.53\%$

C	N	H
$\frac{77.42}{12}$	$\frac{15.05}{14}$	$\frac{7.53}{1}$
6.45	1.075	7.53
$\frac{6.45}{1.075}$	$\frac{1.075}{1.075}$	$\frac{7.53}{1.075}$
6	1	7

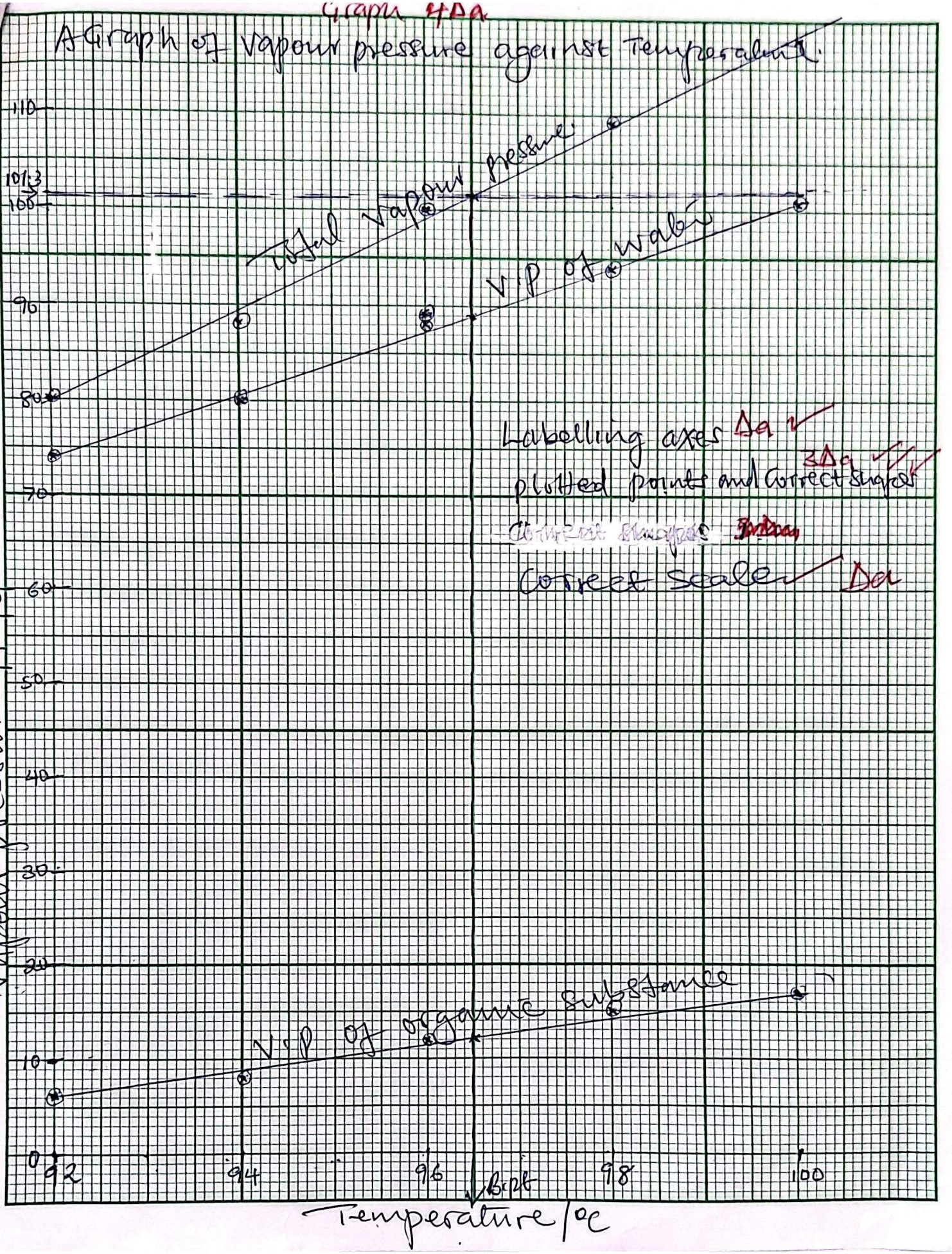
Da

Empirical formula is C_6NH_7

$$(C_6NH_7)_n = 91.78$$
$$((6 \times 12) + 14 + 7)n = 91.78$$
$$93n = 91.78$$

Graph 4Da

A Graph of vapour pressure against Temperature.



Labelling axes Da ✓

plotted points and correct shape 3Da ✓

Correct Scale Da ✓

ITEM 2

Criteria	Scores
L + 4Fe	= 2
L + 3Fe	= 1
Da + Di	= 2
Da	= 1
2Ae	= 2
1Ae	= 1
TOTAL	= [6]

SECTION B

ITEM 3

Beryllium can be used in the project.

Beryllium has the smallest atomic radii and highest melting point. The atoms are closed packed this Any one leads to the delocalised electrons being nearer Pe nuclear charge and experience strong nuclear Pe attraction resulted into stronger metallic bond and giant metallic structure. This leads to electronic equipment Pe being strong and durable. Beryllium has low density hence making light and makes the equipment Pe being portable.

There is increase in atomic radii from beryllium to barium. Electrons are being added different energy levels and there is addition of completely filled energy level with electrons. This leads to increase in Pe screening effect. Protons are also added to the nucleus leading to increase in nuclear charge Pe However the increase in screening effect outweighs increase in nuclear charge resulting into decrease Pe in effective nuclear charge. The outer

electrons therefore become less strongly attracted leading to increase in atomic radii.

Strontium reduces water to hydrogen gas and itself oxidised to Strontium hydroxide

$Sr_{(s)} + 2H_2O_{(l)} \longrightarrow Sr(OH)_{2(s)} + H_{2(g)}$

Accept any one equation with water

Barium reduces water to hydrogen gas and itself oxidised to barium hydroxide.

$Ba_{(s)} + 2H_2O_{(l)} \longrightarrow Ba(OH)_{2(s)} + H_{2(g)}$

Accept one Eq with reactant

Strontium when heated reduces steam to hydrogen gas and itself oxidised to Strontium oxide.

$Sr_{(s)} + H_2O_{(g)} \longrightarrow SrO_{(s)} + H_{2(g)}$

Barium when heated reduces steam to hydrogen gas and itself oxidised to barium oxide

$Ba_{(s)} + H_2O_{(g)} \longrightarrow BaO_{(s)} + H_{2(g)}$

Beryllium reduces warm dilute sulphuric acid to hydrogen gas and itself oxidised to barium sulphate

$Be_{(s)} + H_2SO_{4(aq)} \longrightarrow BeSO_{4(aq)} + H_{2(g)}$

Strontium reduces dilute sulphuric acid to hydrogen gas and itself oxidised to Strontium sulphate, But the reaction occurs for a short time then stops due to formation of insoluble sulphates.

$Sr_{(s)} + H_2SO_{4(aq)} \longrightarrow SrSO_{4(s)} + H_{2(g)}$

Barium reduces dilute sulphuric acid to hydrogen gas and itself oxidised to barium sulphate. However the reaction occurs for a short time then stops due to formation of the insoluble sulphate.

$$Ba_{(s)} + H_2SO_{4(aq)} \rightarrow BaSO_{4(s)} + H_{2(g)}$$

Beryllium reduces concentrated sulphuric acid to sulphur dioxide and water and itself oxidised to beryllium sulphate.

$$Be_{(s)} + 2H_2SO_{4(l)} \rightarrow BeSO_{4(aq)} + SO_{2(g)} + 2H_2O_{(l)}$$

Strontium reduces concentrated sulphuric acid to sulphur dioxide and water and itself oxidised to strontium sulphate. The reaction occurs for a short time then stops due to formation of insoluble sulphate

Accept any one

$$Sr_{(s)} + 2H_2SO_{4(l)} \rightarrow SrSO_{4(s)} + SO_{2(g)} + 2H_2O_{(l)}$$

Barium reduces concentrated sulphuric acid to sulphur dioxide and water and itself oxidised to barium sulphate. The reaction occurs for a short time and then stops due to formation of the insoluble sulphate

$$Ba_{(s)} + 2H_2SO_{4(l)} \rightarrow BaSO_{4(s)} + SO_{2(g)} + 2H_2O_{(l)}$$

As beryllium is used in manufacture of electronic equipment because its a good conductor of electricity for joining as the metal is melted the fumes which are toxic can enter through the mouth or nose which can cause cancer. This can be mitigated by wearing masks to avoid fumes entering through the nose and mouth. Beryllium is used in making of electronic

because its a good conductor of electricity

ITEM 3

Criteria	Scores
7Pe	= 2
(5-6e) Pe	= 1

7Re + 7Eq	= 3
7Re + (5-6) Eq	= 2
(5-6) Eq	= 1

Ea + Ed + Em	= 3
Ea + Ed	= 2
Ea	= 1
TOTAL	8

ITEM 4

They can use carbon because it has smaller atomic radii and higher melting point than iron. Carbon acts as interstitial element when added to iron. This makes the delocalised electrons both from iron and graphite experience strong nuclear attraction. The delocalised electrons will be nearer the nuclear charge and experience strong nuclear attraction. The alloy formed will have stronger metallic bond hence being stronger and harder.

Both Carbon and silicon have giant Covalent structure with strong covalent bonds. However carbon has smaller atomic radii than silicon the bonding electrons in carbon experience stronger nuclear attraction making the covalent bond in carbon being stronger than that in silicon.

Lead has metallic bond and giant metallic structure however the strength of metallic bond in lead is less strong as compared to strength of covalent bonds in carbon and silicon.

Carbon when heated reacts with steam and reduces steam to hydrogen and itself oxidised to carbon monoxide.

$$C_{(s)} + H_2O_{(g)} \longrightarrow CO_{(g)} + H_{2(g)}$$ Eq

Carbon when heated

Silicon when heated reduces steam to hydrogen and itself oxidised to silicon dioxide

$$Si(s) + 2H_2O(g) \longrightarrow SiO_2(s) + 2H_2(g)$$

Lead reacts with water containing dissolved oxygen and is oxidised to Lead(II) hydroxide.

$$2Pb(s) + 2H_2O(l) + O_2(g) \longrightarrow 2Pb(OH)_2(s)$$

Tin when heated is oxidised by chlorine gas to tin(IV) chloride

$$Sn(s) + 2Cl_2(g) \longrightarrow SnCl_4(s)$$

Lead when heated is oxidised by chlorine gas to Lead(II) chloride.

$$Pb(s) + Cl_2(g) \longrightarrow PbCl_2(s)$$

with air/gas, mixture instead of iron oxide Carbon is heated some of the carbon may burn to form carbon dioxide gas which leads to greenhouse effect hence global warming.

Silicon when heated reduces steam to hydrogen and itself oxidised to silicon dioxide

$$Si_{(s)} + 2H_2O_{(g)} \longrightarrow SiO_{2(s)} + 2H_{2(g)}$$

Lead reacts with water containing dissolved oxygen and is oxidised by oxygen to lead(II) hydroxide.

$$2Pb_{(s)} + 2H_2O_{(l)} + O_{2(g)} \longrightarrow 2Pb(OH)_{2(s)}$$

Tin when heated is oxidised by chlorine to tin(IV) chloride

$$Sn_{(s)} + 2Cl_{2(g)} \longrightarrow SnCl_{4(l)}$$

Lead when heated is oxidised by chlorine to Lead(II) chloride.

$$Pb_{(s)} + Cl_{2(g)} \longrightarrow PbCl_{2(s)}$$

When carbon is used the alloy formed is harder and stronger than iron metal however as the mixture is melted during formation of alloy, some of the carbon might burn and form carbondioxide gas which lead to green house effect hence causing global warming. This can be mitigated by planting more trees in order to trap the carbondioxide.

ITEM 4

Criteria	scores
3Pe	= 2
2Pe	= 1

5Eq +5Re	= 3
(3-4) Eq +5Re	= 2
(3-4) Eq	= 1

Ea+Ed+Em	= 3
Ea+Ed	= 2
Ea	= 1
TOTAL	8

ITEM 5

Compound Z is phenylethene used in manufacture of polyphenylethene. Both phenylethene and polyphenylethene posses vander waal forces of attraction. Therefore can dissolve into each other during its preparation. Both polyphenylethene and phenylethene are non polar hence can dissolve into each other.

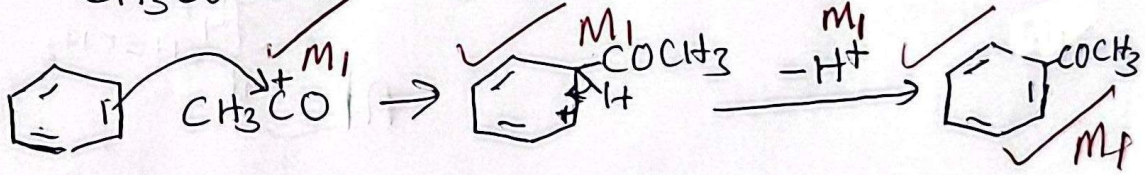
The polyphenylethene is a polymer of high molecular mass and therefore phenylethene with relatively high molecular mass is used.

polyphenylethene is used in making of packaging materials because it is strong, It is non biodegradable. When dumped in the soil will make the soil less productive for use. This can be mitigated by recycling and transform them into some other useful products.

$$C_6H_6 + CH_3COCl \xrightarrow{AlCl_3}$$

Mechanism

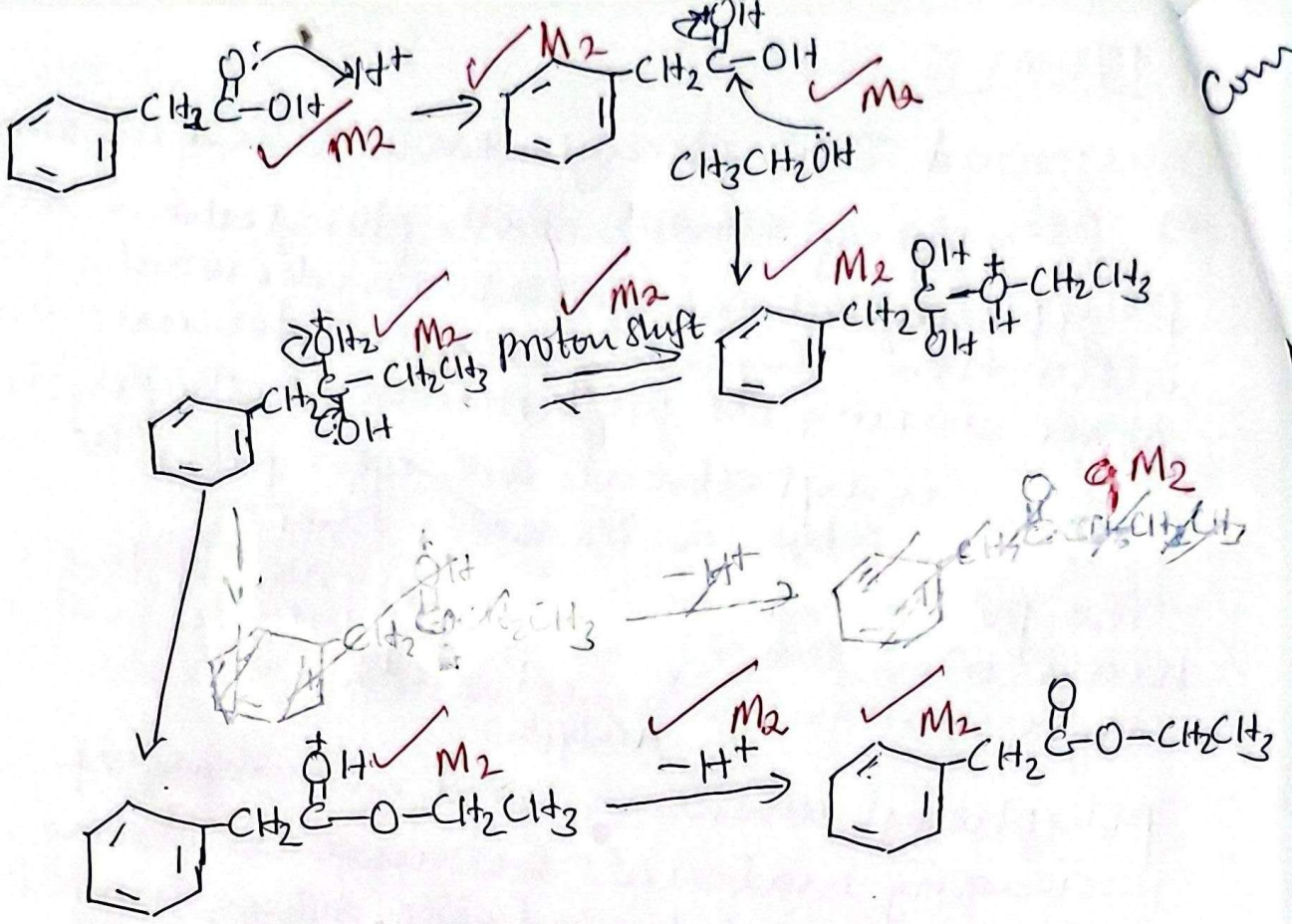
$$CH_3COCl + AlCl_3 \rightleftharpoons CH_3\overset{+}{C}O + AlCl_4^-$$



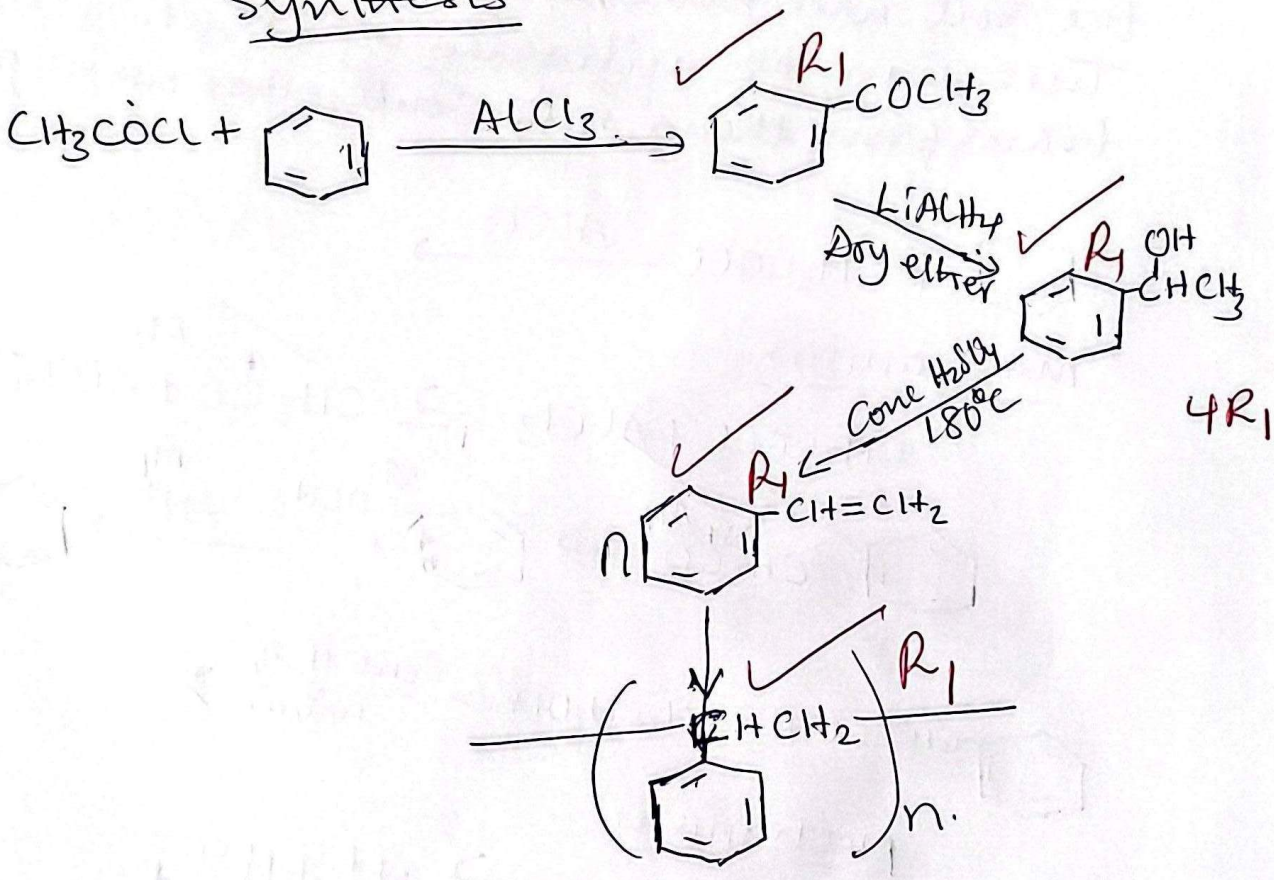
$$CH_2COOH + CH_3CH_2OH \xrightarrow[\text{warm.}]{\text{Conc } H_2SO_4}$$

Mechanism

$$H_2SO_4 \rightleftharpoons H^+ + HSO_4^-$$

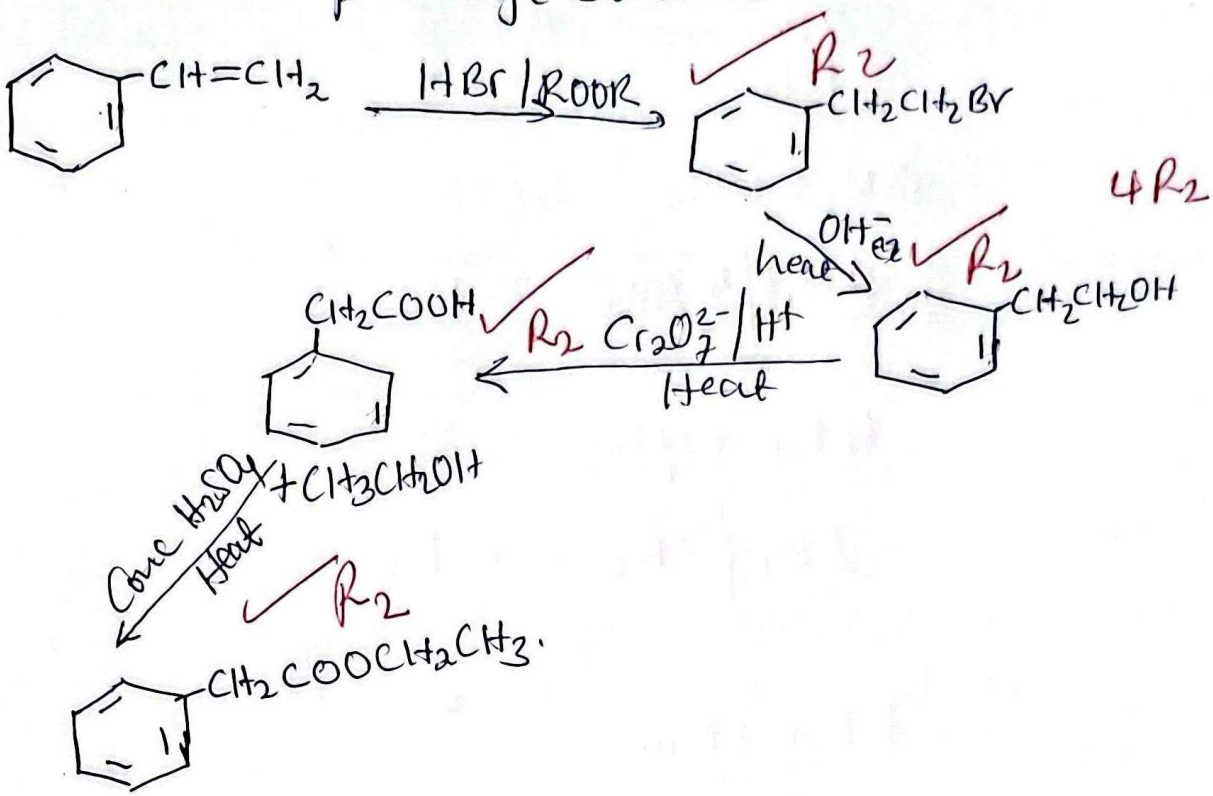


Synthesis



4 R_1

Compound 2 is phenyl ethene



ITEM 5

Criteria		Scores
3Pe	=	2
2Pe	=	1

$5M_1 + 9M_2 = 2$

$(3-4)M_1 + (6-8)M_2 = 1$

$4R_1 + 4R_2 = 2$

$3R_1 + 3R_2 = 1$

$E_a + E_d + E_m = 3$

$E_a + E_d = 2$

$E_a = 1$

TOTAL [9]

ITEM 6

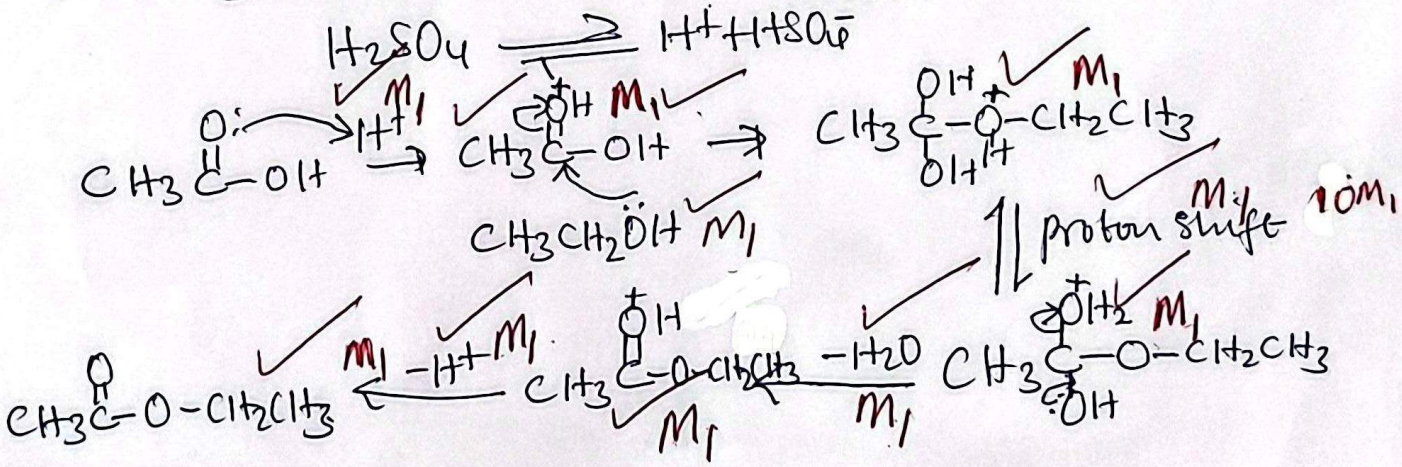
The perfume is volatile and non polar. Therefore compound Q which is ethanol is used because its volatile and less polar solvent. The less polar solvent ethanol enables the perfume dissolve and remain in it during its preparation.

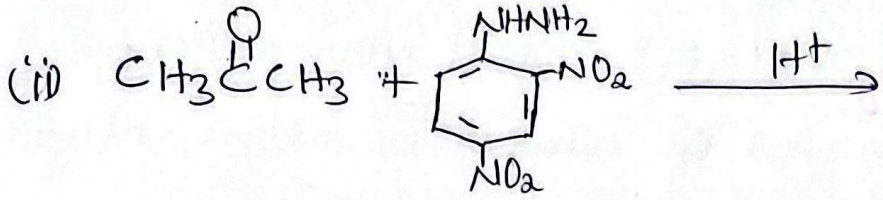
The perfume is a liquid and this makes ethanol suitable for use since its also a liquid can be used as a solvent to dissolve the perfume during its preparation.

Perfumes are used in households because it has good smell. It may cause some respiratory issues during its use to some people who are asthmatic This can be mitigated by people seeking medical consultations before use.

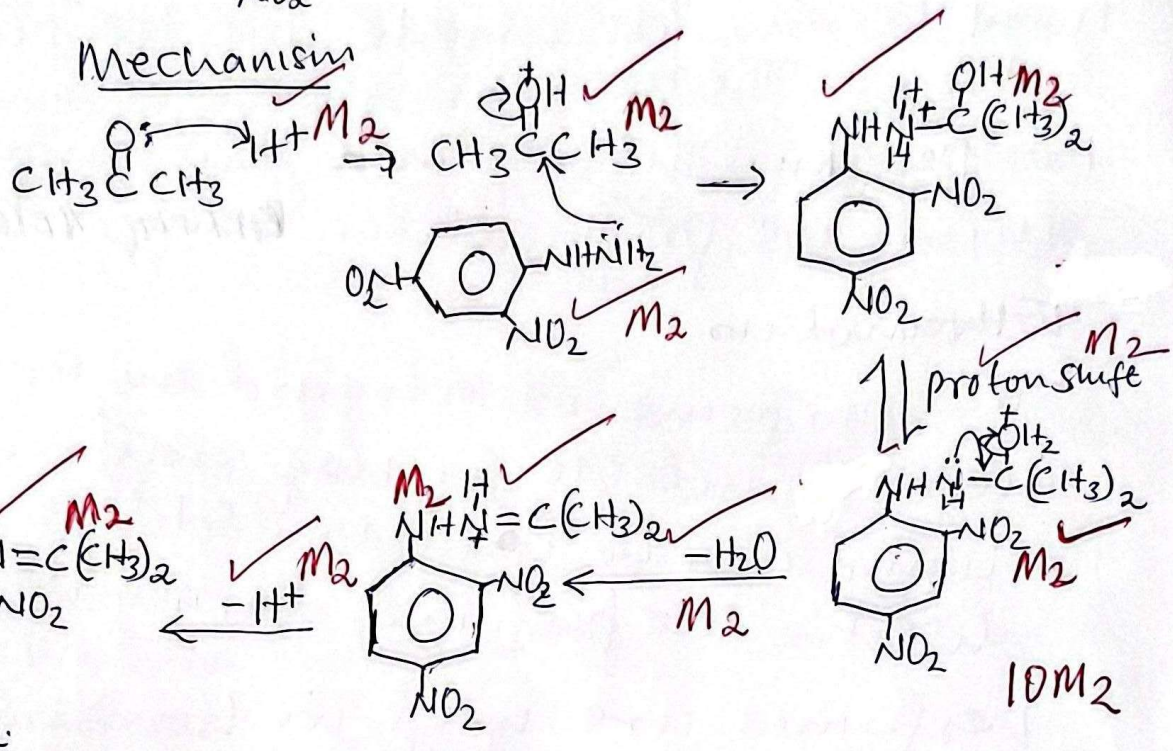
Mechanism

(i) $CH_3CH_2OH + CH_3COOH \xrightarrow[\text{warm}]{\text{Conc } H_2SO_4}$



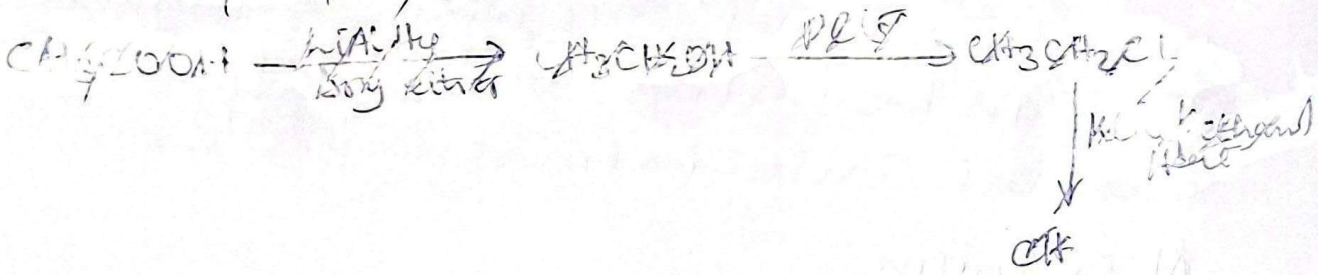
(ii) 

Mechanism



10 M2

Synthesis



(i) $CH_3COOH \xrightarrow[\text{Dry ether}]{LiAlH_4} CH_3CH_2OH \xrightarrow{PCl_5} CH_3CH_2Cl$ R_1

$CH_3CH_2Cl \xrightarrow[\text{Heat}]{KCN/\text{ethanol}} CH_3CH_2CN$ $4R_1$

$CH_3CH_2CN \xrightarrow[\text{Dry ether}]{LiAlH_4} CH_3CH_2CH_2NH_2$ R_1

(ii) $CH_3COOH \xrightarrow{CaCO_3} (CH_3COO)_2Ca \xrightarrow{heat} CH_3COCH_3$ R_2

$CH_3COCH_3 \xrightarrow[\text{Dry ether}]{LiAlH_4} CH_3CH(OH)CH_3$ $6R_2$

$CH_3CH(OH)CH_3 \xrightarrow[180^\circ C]{Conc\ H_2SO_4} CH_3CH=CH_2$ R_2

$CH_3CH=CH_2 \xrightarrow[ROOR]{HBr} CH_3CH_2CH_2Br$ R_2

$CH_3CH_2CH_2Br \xrightarrow[heat]{OH^-_{aq}} CH_3CH_2CH_2OH$ R_2

ITEM 6

Criteria	scores
3 Pe	= 2
2 Pe	= 1

$10M_1 + 10M_2 = 2$

$(7-9)M_1 + (7-9)M_2 = 1$

$4R_1 + 6R_2 = 2$

$3R_1 + (4-5)R_2 = 1$

$Ea + Ed + Em = 3$

$Ea + Ed = 2$

$Ea = 1$

TOTAL 9